GREY HUNTER

SYSTEM ARCHITECTURE & MULTI-PHASE IMPLEMENTATION PLAN

Autonomous Mobile Security Auditing & Defensive Penetration Testing Platform
Official Release Specification • Downloadable Artifact • GitHub Production Ready

1. Executive Summary & Mission Charter

Grey Hunter is an enterprise mobile security platform engineered for ethical hackers, red/purple teams, and defensive penetration testers. The platform unifies native Linux security binaries, hardware-backed Android cryptography, interactive terminal sessions, and automated vulnerability analysis into an intuitive Dragon-themed command suite.

- **Zero-Trust Authentication:** Hardware-backed keystore, biometric session resumption, and RBAC clearance roles.
- **Sandboxed Execution:** Termux PRoot/chroot environment with raw socket privileges (CAP_NET_RAW) and command sanitization.
- **Automated CVSS Telemetry:** Real-time CVE mapping, dynamic score calculations, and defensive remediation playbooks.
- **Encrypted Offline Storage:** SQLCipher 256-bit AES database with zero unencrypted data leakage.

2. System Architecture & Component Topology

Layer	Primary Stack	Core Responsibilities & Security Controls
UI Presentation	React Native 0.73, Redux Toolkit, Dragon UI	Monospace cyber interface, role selection, responsive tablet/mobile terminal views.
Hardware Bridge	Android Keystore, BiometricPrompt, PTY Bridge	Master key generation, StrongBox wrapping, pseudo-terminal session multiplexing.
Execution Sandbox	Termux Runtime, PRoot / Chroot, Raw Sockets	Native tool invocation (Nmap, Netdiscover, Masscan), command sanitization.
Analysis Engine	Streaming Parsers, CVSS 3.1, CVE Correlator	Transform raw terminal stream into structured vulnerability cards and risk grades.
Encrypted DB	SQLCipher, AES-256-GCM, EncryptedStorage	Zero-knowledge local database storing scan reports, targets, and immutable audit logs.
Cloud Bridge	Node.js / Express, Prisma, PostgreSQL, Docker	Multi-operator collaboration, encrypted remote backups, and threat intelligence sync.

3. Detailed 6-Phase Implementation Roadmap

Phase	Focus Area	Deliverables & Key Milestones	Security Controls
Phase 1	**Identity & Auth**	Dragon RegisterScreen, LoginScreen with clearance roles, Keystore integration.	AES-256-GCM, zero-knowledge password hashing, entropy meter.
Phase 2	**Execution Sandboxing**	Netdiscover, Amass, Nuclei, WPScan, RustScan tool wrappers & PTY bridge.	Shell argument sanitization, CAP_NET_RAW checks, scope boundaries.
Phase 3	**Analysis & Telemetry**	Multi-tab ScanResultsScreen, posture scores (A-F), remediation playbooks.	CVSS v3.1 scoring, sensitive client IP data redaction.
Phase 4	**Offline Encrypted DB**	SQLCipher 256-bit database, background sync queue, secure PCAP store.	PBKDF2 64,000 iterations, automated secure memory wipe.
Phase 5	**Collaborative Teaming**	Multi-operator WebSocket mesh, live session tailing, Kanban boards.	mTLS certificate pinning, cryptographically signed audit logs.
Phase 6	**Production Hardening**	ProGuard/R8 obfuscation, root/hook detection, GitHub Actions CI/CD.	OWASP MASVS Level 2, anti-debugging enforcement.

4. Ethical Security Boundaries & Compliance

1. **Pre-Authorized Scope:** Every scan strictly requires pre-approved IP/domain scoping. Out-of-scope probes are halted.
2. **Non-Destructive Operation:** All tools are calibrated for vulnerability inventory and auditing rather than unauthenticated exploitation.
3. **Cryptographic Audit Log:** Each action is sealed with timestamp, target IP, and operator ID into an immutable SQLite log.